

MLFF Training-Data System Contract

Cross-cutting current-generation invariants

mdstats project

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Scope

This document is the cross-cutting current system contract for the mdstats MLFF training-data and fine-tuning workflow. The legacy filename is retained for stable references; this is **not** an implementation-stage plan.

It owns only invariants that span narrower specifications: evidence-role separation, dependency direction, identity/lineage, fitted-domain isolation, target-membership/target-size ownership, protocol identity, replay/monitor separation, sealed evaluation, calibration, bounded execution, and fail-closed current-generation publication.

Narrow specifications own exact module schemas, numerical constants, algorithms, storage formats, and runtime behavior. Architecture owns the higher-level dependency/ownership model. Workplans and historical documents are non-normative.

Normative principles

1. Source facts, eligibility, evidence roles, fitted preparation, target membership, target size, weighting, exposure, checkpoint selection, validation, calibration, and acquisition are distinct record/decision families.

2. A frame that supplied a gradient is not independent validation evidence for that model.
3. Held-out cross-validation evaluates a frozen protocol and cannot control target size, stopping, or checkpoint choice for that protocol.
4. Every fold has a gradient-training domain, an authorized checkpoint monitor, and a held-out evaluation fold with explicit independence/purge evidence.
5. Cross-validation trains a fresh model/optimizer lineage for each held-out fold and validates the complete `TrainingProtocolIdentity` actually used by final training.
6. Feature fitting, E0 fitting, label-derived difficulty evidence, and target-subset inputs inspect only the applicable fold/final gradient-training domain.
7. DATA7 prepares fitted target-subset inputs and SHALL NOT publish target membership or target size.
8. MVSEL2/REPAIR2 are the sole current target-membership ordering/repair authorities; MVSTATE2 is current continuation state; MVQUAL independently verifies hard prefix requirements.
9. `TargetSizeStudyPolicy` is the sole scientific target-size authority. Monitor/replay/batch/pool cardinalities are different semantic types.
10. Target membership is domain-local; the selected target size is protocol-global across required domains.
11. Target-size screening uses only authorized development/model-selection evidence. Held-out CV, calibration, and locked tests are forbidden inputs.
12. Locked tests cannot affect fitting, membership, size, protocol choice, stopping, checkpointing, calibration-policy choice, or acquisition and are activated only after protocol/committee freeze.
13. Replay training, replay monitoring, target monitoring, and target training preserve separate source/role identities.
14. Replay retention and mandatory physical/deployment integrity are hard admissibility constraints unless an explicit current scientific policy states otherwise.
15. Dynamic resampling/exposure semantics require an explicit current adapter/protocol; static files alone cannot claim them.
16. Calibration is bound to predictions from the actual frozen final committee and an explicit applicability domain.
17. Active-learning child generations inherit prior evidence roles unless a new evaluation lineage explicitly reassigns them.
18. Unsupported old campaign generations are rejected/re-prepared rather than migrated into current semantics.
19. Execution caches, worker scheduling, out-of-core layout, and other realization choices cannot change scientific identity or authoritative decisions.
20. Publication fails closed when required current-generation identities, upstream evidence, or schema/content validation are missing/incompatible.

Core record ownership

Record / policy family	Owns		Must not own
TrainingDataSource / source records	source position/label-domain	bytes/controls/composition lin- age	frame eligibility or evidence role
TrainingFrameRecord	immutable	source-bound frame facts	eligibility, partition, membership, exposure
FrameEligibilityDecision	post-label/quality	eligibility	partition or target membership
PartitionAssignment	one statistical role	under DATA5 policy	fitted quantities or target order
PartitionFeasibilityReport	whether requested roles are supportable	evidence	fabricated independent evidence
PartitionIndependenceReport	actual independence/purge/duplicate limitations		stronger independence than observed
DATA6/7 fitted records	training-domain descriptors/transforms/E0/difficulty/objective/weights/subset inputs		held-out labels, target membership, target size
MVSEL2 result	deterministic within one training domain	target order	independent qualification or selected target size
REPAIR2 result	one repaired master domain	order per domain	independently repaired rung copies
MVSTATE2	authenticated selector/repair state	reconstructible continuation	migration semantics for obsolete state
MVQUAL result	independent obligation evidence for a prefix	hard coverage/	selector score/ranking or target-size choice by itself
TargetSizeStudyPolicy / TargetSizeDecision	nominal/materializable/qualified population, fidelity funnel, selected target size or typed failure		monitor construction or held-out CV evaluation
OnlineTargetMonitorPolicy	common target-monitor evidence set		target-training size
ReplayMonitorPolicy	replay-monitor evidence set		target-training size or replay-training membership
TrainingProtocolIdentity	complete frozen model/data/replay/membership/size/objective/exposure/checkpoint/parallelism/runtime protocol identifier and/or probability evidence		mutable runtime observations or test results
CardinalityEstimationEvidence	CardinalityEstimationEvidence	CardinalityEstimationEvidence	CardinalityEstimationEvidence

Every serialized current record SHALL carry a versioned schema, deterministic content identity, explicit upstream lineage, and explicit policy/failure identities as appropriate.

Identity and leakage contract

Source occurrence, geometry, and labels

Source occurrence (`frame_uid` or current equivalent), geometry fingerprint, label payload digest, and combined labeled-configuration fingerprint are distinct identities.

Geometry identity excludes energy/force/stress labels. Label identity includes selected labels and label-domain identity. Leakage audits use exact occurrence overlap, exact geometry overlap, exact labeled-configuration overlap, near-duplicate evidence where required, and forbidden temporal proximity.

Label-domain compatibility

Label-domain identity separates theory/electronic-structure identity, energy-reference identity, derivative/stress convention, numerical-quality profile, and software provenance. A current compatibility policy may accept non-semantic provenance differences but cannot silently merge incompatible theory or energy-reference domains.

One target MACE bundle contains one compatible target label domain plus a separately identified replay lineage where replay is enabled.

Fitted-domain isolation

Raw physical/structural/event facts may be constructed before partitioning when the owning provider is partition-independent. Any learned/fitted transform—including scaling, PCA/whitening, fitted metrics, E0 corrections, or label-derived residual difficulty—is bound to a specific authorized gradient-training domain.

For fold k , the allowed direction is:

```
DATA5 fold_training_domain_k
-> DATA6/7 fitted products
-> MVSEL2/REPAIR2 membership inside that domain
-> selected-size prefix after target-size freeze
-> training/checkpoint choice using authorized monitor
-> held_out_evaluation_fold_k only after checkpoint freeze
```

A reverse dependency from held-out evaluation into fitted products, target size, or checkpoint selection is prohibited.

Target membership and target-size contract

The current target-subset construction chain is:

```
DATA7 TargetSubsetInputBundle
-> FEAS1
-> MVIDX1
```

- > MVSEL2
- > REPAIR2 / MVSTATE2
- > MVQUAL
- > TargetSizeStudyPolicy

No current alternate v1/migration/rescue branch exists.

The exact fixed nominal target-size population and configurable $(n_1, n_2, n_3)/n$ fidelity policy are owned by `mlff_target_subset_size_study_spec.md` and are not duplicated here.

For each required training domain d , candidate membership at size N is the prefix of that domain's one repaired master order. One selected N is frozen into the complete training protocol across required fold/final domains.

Hard prefix qualification is independently owned by MVQUAL. Non-monotone hard qualification over nested increasing prefixes is an invariant failure.

Training-protocol and checkpoint contract

TrainingProtocolIdentity SHALL bind, as applicable:

- foundation/model/head identity
- selected target size and domain-local membership identity
- replay source/training/monitor identities
- common target-monitor identity
- objective and configuration/property weights
- exposure backend and realized balancing/duplication policy
- checkpoint metric and replay-retention policy
- optimizer/LR/stopping/epoch policy
- seed policy
- precision/backend
- MACE adapter/runtime lock

A comparison or CV claim applies only to the protocol identity actually evaluated.

Checkpoint selection uses explicit target/focus/replay/property/integrity constraints. A candidate violating a mandatory constraint is inadmissible even if another target metric is lower.

MACE realization and exposure

Current MACE artifacts contain only supported labels/weights/compact identities in their interchange format; extended provenance remains sidecar/content-addressed.

The adapter verifies current upstream behaviors on which protocol semantics depend: head ordering, loader realization, checkpoint retention/control, precision/backend, and effective target/replay exposure.

Intended exposure cannot substitute for realized exposure. Silent loader duplication or changed target/replay counts fail closed unless the accepted current protocol explicitly binds that behavior.

Runtime/package locks are owned by the narrow current runtime specification and may evolve independently of this cross-cutting contract.

Sealed evaluation, calibration, and active learning

Development, calibration, and locked evaluation artifacts remain role-separated. Locked-test configuration/path access is absent from development control flow until explicit activation after protocol/committee freeze.

Calibration numerical thresholds derive from the actual frozen final committee and a dedicated authorized calibration cohort. Applicability/transfer decisions explicitly distinguish within-domain, rank-only, recalibration-required, and incompatible-domain behavior.

Active-learning labels create a new development generation. Existing role assignments are inherited by default; repartitioning previously classified evidence creates a new evaluation lineage.

Bounded execution and persistence

Scientific policy must be realizable without duplicating product-scale state per target-size rung. The current architectural materialization is one fitted-input authority, one exact sparse neighborhood/MVIDX authority, one repaired master order per training domain, prefix metadata, required MVQUAL evidence, and only currently authorized training artifacts.

Persistent execution caches are reconstructible unless another current specification explicitly makes them scientific evidence. Every cache validates semantic inputs and payload integrity. Corrupt/stale state is rebuilt or fails cleanly; it never changes policy to rescue a run.

Worker count, queue ordering, chunking, file-backed versus in-memory layout, and cache path are non-semantic under an exact-equivalence contract.

Current-generation publication and failure rules

Current products SHALL fail closed when required source/label identity, evidence roles, fitted-domain lineage, selector/repair/qualification identity, target-size decision, replay/monitor lineage, training protocol, runtime behavior, or publication payload validation is missing/incompatible.

Unsupported historical campaign schemas are not current compatibility obligations. Current code may retain low-level readers for forensic purposes, but those readers cannot create a second product-semantic path and are not normative documentation authority.

Extension rule

A new feature/provider may enrich raw or DATA7 inputs without creating another membership selector. A new selector objective, target-size population, evidence role, loss function, stopping rule, or compatibility generation changes scientific protocol semantics and requires explicit architecture/specification revision plus qualification.